Roma Tre

DIPARTIMENTO DI MATEMATICA E FISICA

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Exponential decay of correlations in one-dimensional $H^{2|2n}$ nonlinear sigma models

Speaker: Javier Durán Fernández - University of Bonn

Time: Tuesday November 24, 16:00-16:45

Location: Room B308, Largo S. Murialdo 1 (Roma)

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The $H^{2|2n}$ model generalizes the $H^{2|2}$ model originally introduced by Zirnbauer to study disordered systems. Surprisingly these models are closely related to probabilistic models such as vertex-reinforced jump processes and the arboreal gas. These connections provide further motivation for investigating the decay of correlation functions.

In this talk we establish exponential decay of correlations in one dimension in some simple cases and compare the resulting rates of decay with those obtained in previous work in the high temperature regime. To determine the optimal rate we introduce the necessary tools from Grassmann calculus and define a transfer operator acting on a Grassmann algebra. Exploiting the hyperbolic symmetry of the model we can diagonalize the operator and explicitly compute the eigenvalues to deduce the desired decay.

Based on ongoing joint work with Luca Fresta.